
Session 02 (AIML) - Assignment Questions

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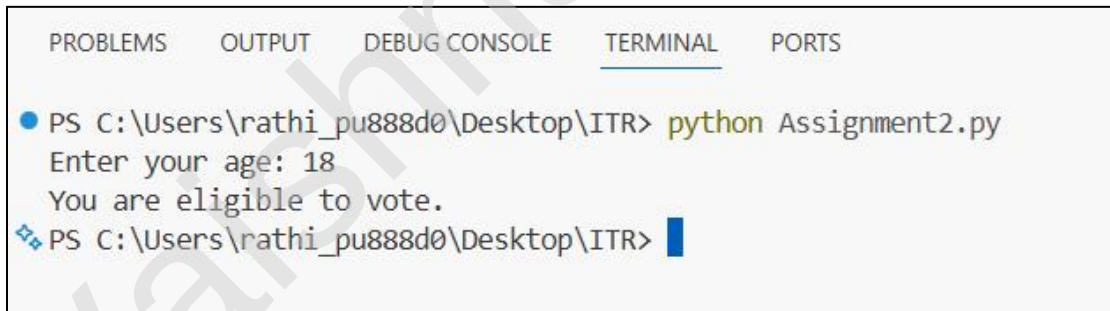
College: **Zeal Polytechnic, Narhe, Pune**

Branch: **Artificial Intelligence and Machine Learning (AIML)**

Domain: **AIML (Artificial Intelligence and Machine Learning)**

Q1. If statement

Output Screenshot:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

● PS C:\Users\rathi_pu888d0\Desktop\ITR> python Assignment2.py
Enter your age: 18
You are eligible to vote.
PS C:\Users\rathi_pu888d0\Desktop\ITR> 
```

Program Code:

```
""" ASSIGNMENT 2 """
```

```
'''
```

Q1. (if Statement)

Write a program that asks the user to enter their age.

If the age is 18 or more, print the message:

"You are eligible to vote."

(Do not print anything if the condition is False).

Add comments.

```
'''
```

```
# Ask the user to enter their age
```

```
age = int(input("Enter your age: "))
```

```
# Check if the age is 18 or more
```

```
if age >= 18:
```

```
    print("You are eligible to vote.")
```

```
'''Explanation in Comments'''
```

```
# The variable 'age' stores the age entered by the user.
```

```
# The if statement checks whether the age is greater than or equal to 18.
```

```
# If the condition is True, the program prints:
```

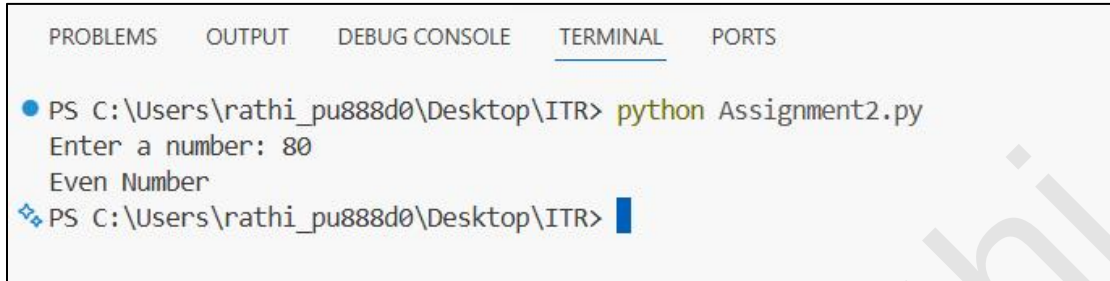
```
# "You are eligible to vote."
```

```
# If the condition is False, nothing is printed.
```

```
# The variable 'is_Student' is a boolean that indicates whether I am a student or not
```

Q2. if-else statements

Output Screenshot:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\rathi_pu888d0\Desktop\ITR> python Assignment2.py
Enter a number: 80
Even Number
PS C:\Users\rathi_pu888d0\Desktop\ITR>
```

Program Code:

```
""" ASSIGNMENT 2 """
```

```
'''
```

Q2. (if-else Statement)

Write a program that takes a number as input from the user and checks whether it is even or odd.

Print "Even Number" or "Odd Number" accordingly.

```
'''
```

```
# Ask the user to enter a number
```

```
number = int(input("Enter a number: "))
```

```
# Check whether the number is even or odd
```

```
if number % 2 == 0:
```

```
    print("Even Number")
```

```
else:
```

```
    print("Odd Number")
```

```
"""Explanation in Comments"""
```

```
# The variable 'number' stores the number entered by the user.
```

```
# The modulus operator (%) is used to find the remainder when dividing by 2.
```

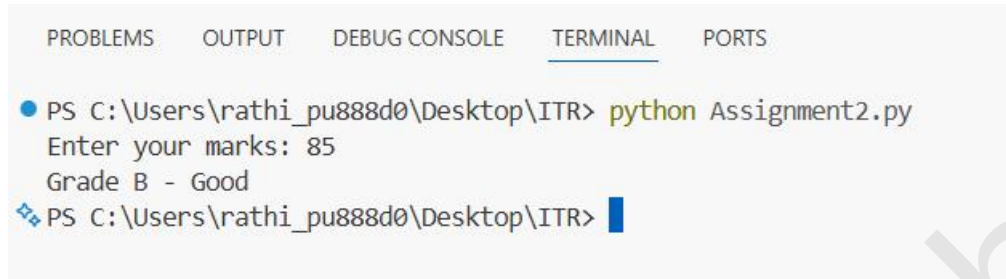
```
# If the remainder is 0, the number is even.
```

```
# Otherwise, the number is odd.
```

```
# The if-else statement is used to print the appropriate message.
```

Q3. if-elif-else statements

Output Screenshot:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\rathi_pu888d0\Desktop\ITR> python Assignment2.py
Enter your marks: 85
Grade B - Good
PS C:\Users\rathi_pu888d0\Desktop\ITR>
```

Program Code:

```
""" ASSIGNMENT 2 """
```

```
'''
```

Q3. (if-elif-else Statement)

Write a program to assign grades based on marks entered by the user:

Marks $\geq 90 \rightarrow$ "Grade A - Excellent"

Marks $\geq 75 \rightarrow$ "Grade B - Good"

Marks $\geq 60 \rightarrow$ "Grade C - Average"

Marks $\geq 40 \rightarrow$ "Grade D - Pass"

Marks $< 40 \rightarrow$ "Fail"

Use if-elif-else and print the appropriate grade message.

```
'''
```

```
marks = int(input("Enter your marks: "))
```

```
if marks >= 90:
```

```
    print("Grade A - Excellent")
```

```
elif marks >= 75:
```

```
    print("Grade B - Good")
```

```
elif marks >= 60:
```

```
    print("Grade C - Average")
```

```
elif marks >= 40:
```

```
    print("Grade D - Pass")
```

```
else:
```

```
    print("Fail")
```

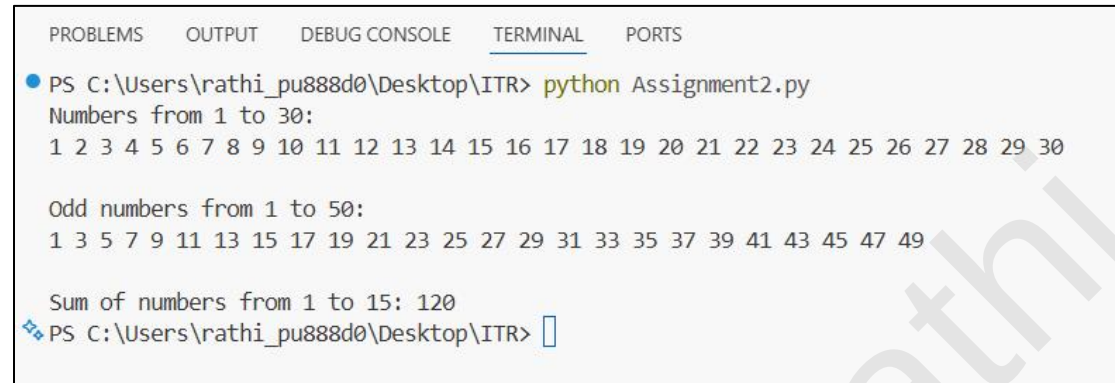
""Explanation in Comments""

```
# The variable 'marks' stores the marks entered by the user.  
# The if-elif-else statement is used to check different ranges of marks.  
# If marks are 90 or above, Grade A is assigned.  
# If marks are between 75 and 89, Grade B is assigned.  
# If marks are between 60 and 74, Grade C is assigned.  
# If marks are between 40 and 59, Grade D is assigned.  
# If marks are below 40, the student is declared Fail.
```

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Q4. For Loop with range

Output Screenshot:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
PS C:\Users\rathi_pu888d0\Desktop\ITR> python Assignment2.py
Numbers from 1 to 30:
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30

Odd numbers from 1 to 50:
1 3 5 7 9 11 13 15 17 19 21 23 25 27 29 31 33 35 37 39 41 43 45 47 49

Sum of numbers from 1 to 15: 120
PS C:\Users\rathi_pu888d0\Desktop\ITR> 
```

Program Code:

```
""" ASSIGNMENT 2 """
```

```
'''
```

Q4. (For Loop with range)

Write a program using for loop and range() to:

a) Print all numbers from 1 to 30.

b) Print all odd numbers from 1 to 50.

c) Print the sum of numbers from 1 to 15.

Add comments explaining the use of range().

```
'''
```

```
# a) Print all numbers from 1 to 30
```

```
print("Numbers from 1 to 30:")
```

```
for i in range(1, 31):
```

```
    print(i, end=' ') # end=' ' is used to print numbers in the same line separated by space
```

```
print("\n")
```

```
# b) Print all odd numbers from 1 to 50
```

```
print("Odd numbers from 1 to 50:")
```

```
for i in range(1, 51, 2):
```

```
    print(i, end=' ') # end=' ' is used to print numbers in the same line separated by space
```

```
print("\n")
```

```
# c) Print the sum of numbers from 1 to 15
sum = 0
```

```
for i in range(1, 16):
    sum += i
```

```
print("Sum of numbers from 1 to 15:", sum)
```

"Explanation in Comments"

range(start, stop) generates numbers from start to stop-1.

range(1, 31) generates numbers from 1 to 30.

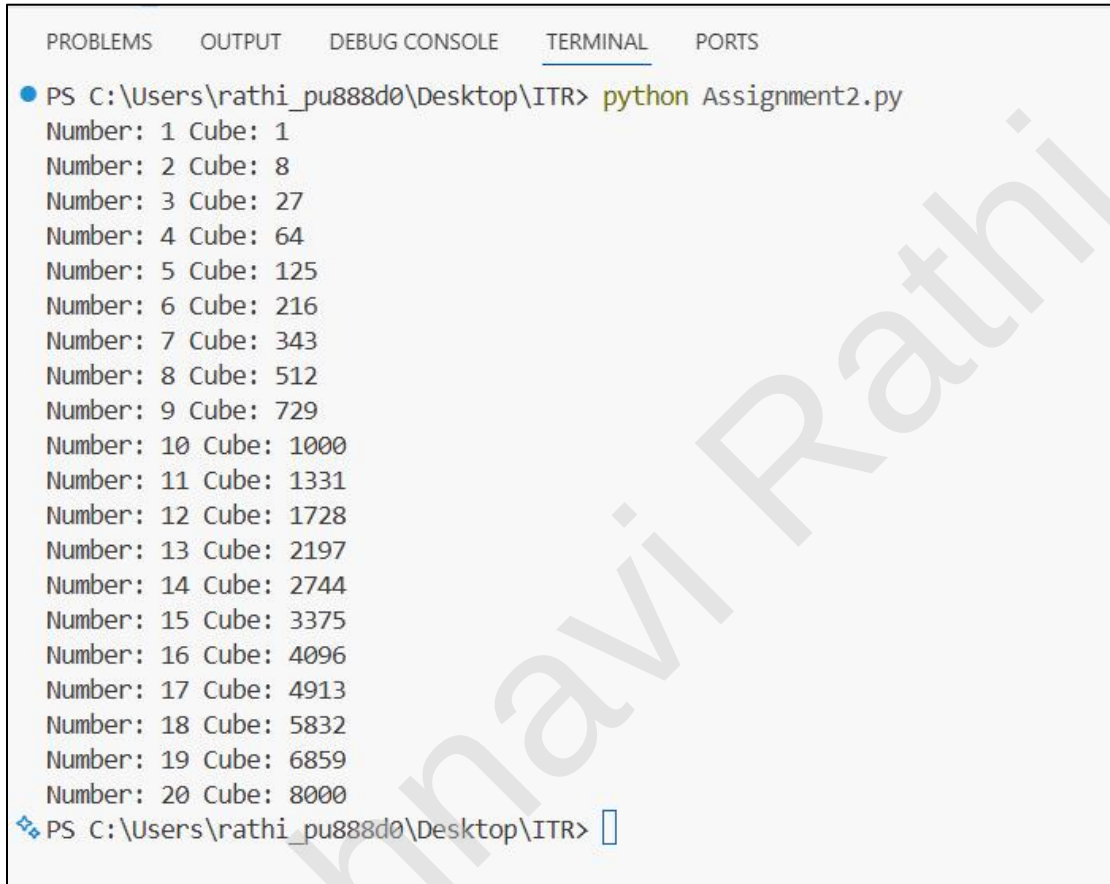
range(1, 51, 2) generates odd numbers from 1 to 49 by increasing by 2.

The for loop is used to iterate through the values generated by range().

In part (c), the variable 'sum' stores the running total of numbers from 1 to 15.

Q5. While Loop

Output Screenshot:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\rathi_pu888d0\Desktop\ITR> python Assignment2.py
Number: 1 Cube: 1
Number: 2 Cube: 8
Number: 3 Cube: 27
Number: 4 Cube: 64
Number: 5 Cube: 125
Number: 6 Cube: 216
Number: 7 Cube: 343
Number: 8 Cube: 512
Number: 9 Cube: 729
Number: 10 Cube: 1000
Number: 11 Cube: 1331
Number: 12 Cube: 1728
Number: 13 Cube: 2197
Number: 14 Cube: 2744
Number: 15 Cube: 3375
Number: 16 Cube: 4096
Number: 17 Cube: 4913
Number: 18 Cube: 5832
Number: 19 Cube: 6859
Number: 20 Cube: 8000
PS C:\Users\rathi_pu888d0\Desktop\ITR> 
```

Program Code:

```
""" ASSIGNMENT 2 """
```

```
'''
```

Q5. (While Loop)

Write a program using while loop to print the following:
Numbers from 1 to 20.

Also print the cube of each number (i.e., 1^3 , 2^3 , 3^3 , ...).

Stop when the number exceeds 20.

```
'''
```

```
# Initialize the number with 1
```



```
num = 1
```

```
# Continue the loop until the number becomes greater than 20
```

```
while num <= 20:
```

```
    # Print the number and its cube
```

```
    print("Number:", num, "Cube:", num ** 3)
```

```
    # Increase the value of num by 1
```

```
    num += 1
```

```
"""Explanation in Comments"""
```

```
# The variable 'num' is initialized with the value 1.
```

```
# The while loop runs as long as num is less than or equal to 20.
```

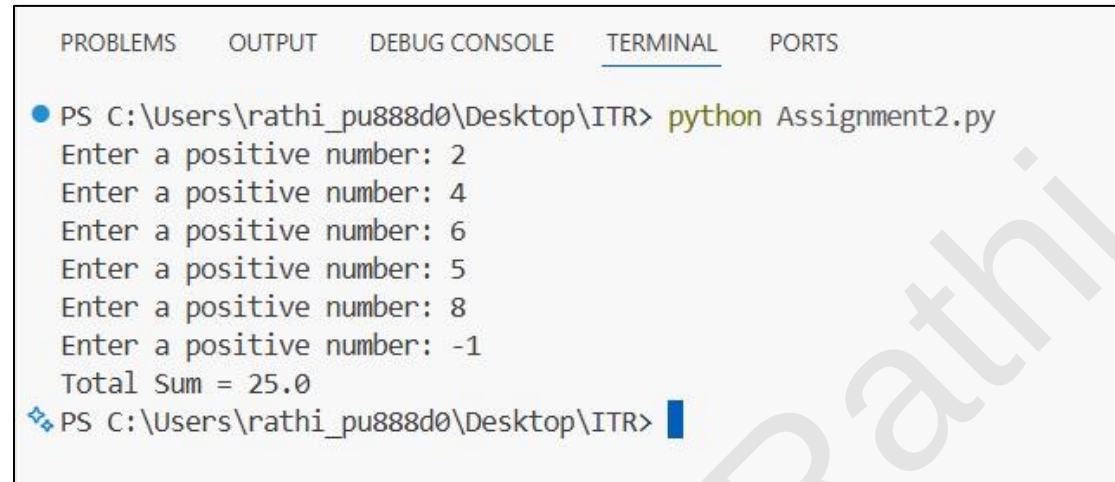
```
# The expression num ** 3 calculates the cube of the number.
```

```
# After each iteration, num is increased by 1.
```

```
# The loop automatically stops when num becomes greater than 20.
```

Q6. While Loop - User Controlled

Output Screenshot:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\rathi_pu888d0\Desktop\ITR> python Assignment2.py
Enter a positive number: 2
Enter a positive number: 4
Enter a positive number: 6
Enter a positive number: 5
Enter a positive number: 8
Enter a positive number: -1
Total Sum = 25.0
PS C:\Users\rathi_pu888d0\Desktop\ITR>
```

Program Code:

```
""" ASSIGNMENT 2 """

"""
Q6. (While Loop - User Controlled)
Write a program that repeatedly asks the user to enter a positive number.
Keep adding the entered numbers to a running total.
Stop when the user enters a negative number or zero.
Finally, print the total sum.
"""

# Initialize the total sum
total = 0

# Ask the user to enter a number
number = float(input("Enter a positive number: "))

# Continue the loop while the number is positive
while number > 0:

    # Add the entered number to the total
    total += number

    # Ask for another number
    number = float(input("Enter a positive number: "))

# Display the final sum
```

```
print("Total Sum =", total)
```

```
"""Explanation in Comments"""
```

```
# The variable 'total' stores the running sum of positive numbers.
```

```
# The user is asked to enter a number.
```

```
# The while loop continues as long as the entered number is greater than 0.
```

```
# Each positive number is added to 'total'.
```

```
# If the user enters 0 or a negative number, the loop stops.
```

```
# Finally, the program prints the total sum of all positive numbers entered.
```

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Q7. Nested-if Statement

Output Screenshot:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\rathi_pu888d0\Desktop\ITR> python Assignment2.py
Enter temperature: 29
Is it raining? (yes/no): no
Weather is pleasant. Have a nice day!
PS C:\Users\rathi_pu888d0\Desktop\ITR>
```

Program Code:

```
""" ASSIGNMENT 2 """
```

```
'''
```

Q7. (Nested if Statement)

Write a program that takes two inputs:

temperature (integer)

is_raining (enter "yes" or "no")

Then give advice using nested if:

If temperature > 30 and not raining → "Hot day, wear light clothes."

If temperature > 30 and raining → "Hot and rainy, carry umbrella."

If temperature < 15 and raining → "Cold and rainy, wear jacket and take umbrella."

Else give a suitable message.

```
'''
```

```
# Take input from the user
```

```
temperature = int(input("Enter temperature: "))
```

```
is_raining = input("Is it raining? (yes/no): ")
```

```
# Check weather conditions using nested if
```

```
if temperature > 30:
```

```
    if is_raining == "yes":
```

```
        print("Hot and rainy, carry umbrella.")
```

```
    else:
```

```
        print("Hot day, wear light clothes.")
```

```
elif temperature < 15:
```

```
if is_raining == "yes":  
    print("Cold and rainy, wear jacket and take umbrella.")  
else:  
    print("Cold day, wear warm clothes.")
```

```
else:  
    print("Weather is pleasant. Have a nice day!")
```

"Explanation in Comments"

```
# The variable 'temperature' stores the temperature entered by the user.  
# The variable 'is_raining' stores whether it is raining or not.  
# A nested if statement means an if statement inside another if statement.  
# If the temperature is greater than 30, the program checks whether it is raining.  
# If the temperature is less than 15, the program again checks whether it is raining.  
# Depending on the conditions, the appropriate advice is displayed.  
# If none of the specified conditions are met, a general message is printed.
```

Q8. For Loop + if-elif-else

Output Screenshot:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\rathi_pu888d0\Desktop\ITR> python Assignment2.py
1
2
Fizz 4
Buzz Fizz 7
8
Fizz Buzz 11
Fizz 13
14
FizzBuzz 16
17
Fizz 19
Buzz Fizz 22
23
Fizz Buzz 26
Fizz 28
29
FizzBuzz 31
32
Fizz 34
Buzz Fizz 37
38
Fizz Buzz
PS C:\Users\rathi_pu888d0\Desktop\ITR>
```

Program Code:

```
""" ASSIGNMENT 2 """
```

```
'''
```

```
Q8. (For Loop + if-elif-else)
```

Write a program using for loop that goes from 1 to 40 and:

Prints "Fizz" if number is divisible by 3

Prints "Buzz" if number is divisible by 5

Prints "FizzBuzz" if divisible by both 3 and 5

Otherwise prints the number

```
'''
```

```
# Loop through numbers from 1 to 40
```

```
for num in range(1, 41):

    # Check if the number is divisible by both 3 and 5
    if num % 3 == 0 and num % 5 == 0:
        print("FizzBuzz",end=" ")

    # Check if the number is divisible by 3
    elif num % 3 == 0:
        print("Fizz",end=" ")

    # Check if the number is divisible by 5
    elif num % 5 == 0:
        print("Buzz",end=" ")

    # Print the number if none of the above conditions are true
    else:
        print(num)
```

"Explanation in Comments"

```
# The for loop iterates through numbers from 1 to 40.
# The modulus operator (%) is used to check divisibility.
# If a number is divisible by both 3 and 5, "FizzBuzz" is printed.
# If a number is divisible only by 3, "Fizz" is printed.
# If a number is divisible only by 5, "Buzz" is printed.
# Otherwise, the number itself is printed.
# The condition for "FizzBuzz" is checked first to avoid missing numbers divisible by
both 3 and 5.
and 5.
```

Q9.Menu Driven Program

Output Screenshot:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\rathi_pu888d0\Desktop\ITR> python Assignment2.py

Simple Calculator
1. Add two numbers
2. Subtract two numbers
3. Multiply two numbers
4. Divide two numbers
5. Exit
Enter your choice: 1
Enter first number: 10
Enter second number: 5
Result = 15.0

Simple Calculator
1. Add two numbers
2. Subtract two numbers
3. Multiply two numbers
4. Divide two numbers
5. Exit
Enter your choice: 5
Exiting Calculator...
PS C:\Users\rathi_pu888d0\Desktop\ITR>
```

Program Code:

```
""" ASSIGNMENT 2 """
```

```
'''
```

Q9. (Menu Driven Program)

Create a simple calculator using while loop and if-elif-else.

Display this menu repeatedly:

1. Add two numbers
2. Subtract two numbers
3. Multiply two numbers
4. Divide two numbers
5. Exit

Ask user for choice.
Take two numbers as input.
Perform the operation.
Continue until user chooses 5 (Exit).
Handle division by zero case.
"""

Run the program until the user chooses Exit
while True:

 # Display menu

 print("\nSimple Calculator")
 print("1. Add two numbers")
 print("2. Subtract two numbers")
 print("3. Multiply two numbers")
 print("4. Divide two numbers")
 print("5. Exit")

 # Take user's choice

 choice = int(input("Enter your choice: "))

 # Exit the program if choice is 5

 if choice == 5:
 print("Exiting Calculator...")
 break

 # Take two numbers as input

 num1 = float(input("Enter first number: "))
 num2 = float(input("Enter second number: "))

 # Perform operation based on user's choice

 if choice == 1:
 print("Result =", num1 + num2)

 elif choice == 2:
 print("Result =", num1 - num2)

 elif choice == 3:
 print("Result =", num1 * num2)

 elif choice == 4:
 if num2 != 0:
 print("Result =", num1 / num2)
 else:
 print("Error: Division by zero is not allowed.")

 else:
 print("Invalid Choice. Please try again.")

"""Explanation in Comments"""

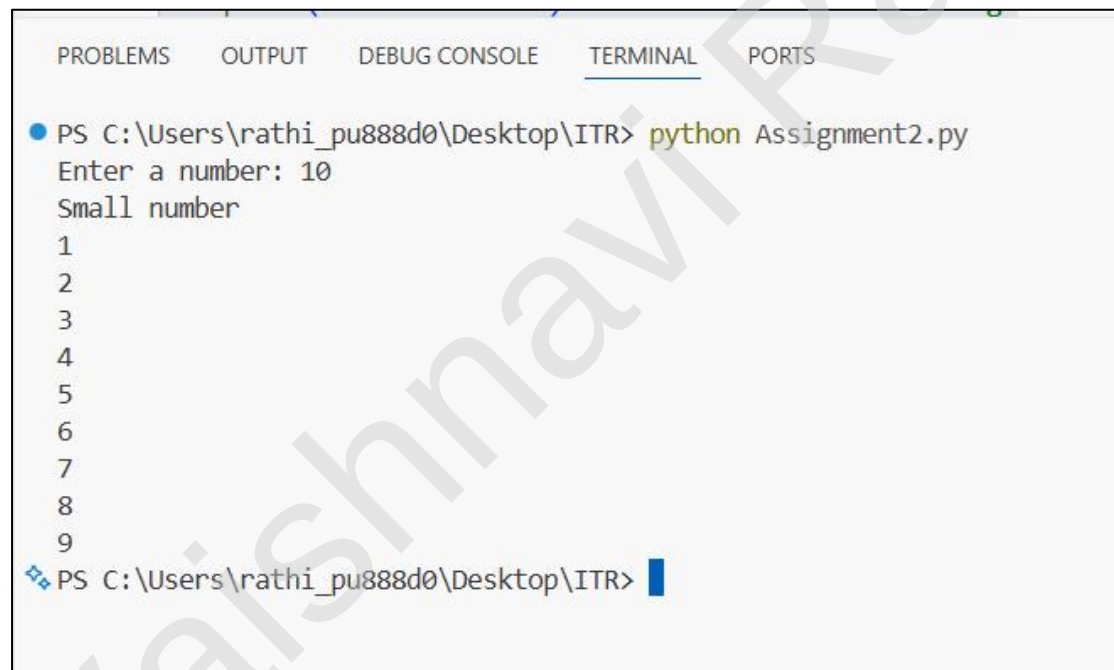
The while True loop keeps displaying the menu until the user selects Exit.
The variable 'choice' stores the user's selected operation.
Two numbers are taken as input for the calculation.
if-elif-else statements are used to perform different operations.
The break statement terminates the loop when the user chooses 5.
Before division, the program checks whether the second number is zero.
This prevents division by zero errors.

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Q10. Debugging

```
num = input("Enter a number: ")
if num > 100
print("Large number")
elif num > 50
print("Medium number")
else
print("Small number")
count = 1
while count < 10
print(count)
count + 1
```

Output Screenshot:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

● PS C:\Users\rathi_pu888d0\Desktop\ITR> python Assignment2.py
Enter a number: 10
Small number
1
2
3
4
5
6
7
8
9
PS C:\Users\rathi_pu888d0\Desktop\ITR>
```

Fixed Program Code:

```
""" ASSIGNMENT 2 """
```

```
'''
```

```
Q10. (Debugging Program)
```

The following code contains several errors.

Correct the program and add meaningful comments explaining each fix.

```
'''
```

```
# input() returns a string, so convert it to an integer for comparison
```

```
num = int(input("Enter a number: "))
```

```
if num > 100: # Missing colon (:)
```

```
    print("Large number") # Indentation was missing
```

```
elif num > 50: # Missing colon (:)
```

```
    print("Medium number") # Indentation was missing
```

```
else: # Missing colon (:)
```

```
    print("Small number") # Indentation was missing
```

```
count = 1
```

```
while count < 10: # Missing colon (:)
```

```
    print(count) # Indentation was missing
```

```
    count += 1 # count + 1 does not update the value of count
```
